

Keivan Faghih Niresi

Machine Learning and Signal Processing Researcher | Ph.D. Candidate at EPFL

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Research Interests

Machine Learning Graph neural networks, Uncertainty quantification, Domain adaptation, Physics-informed learning

Signal Processing Computational sensing/imaging, Inverse problems, Graph signal processing, High-dimensional data analysis

Education

École Polytechnique Fédérale de Lausanne (EPFL)

Docteur ès Sciences (Ph.D.)

Lausanne, CH

Feb. 2023 - Dec. 2026 (Expected)

- **Thesis:** *Computational Sensing over Graphs and Time for Engineered Systems*
- **Advisor:** Prof. Olga Fink

National Tsing Hua University (NTHU)

Master of Science (M.Sc.) in Communications Engineering

Hsinchu, TW

Sep. 2020 - Nov. 2022

- **Thesis:** *Hyperspectral Image Restoration Framework Based on Robust Untrained Neural Networks*
- **Advisor:** Prof. Chong-Yung Chi

University of Guilan

Bachelor of Science (B.Sc.) in Electrical Engineering

Rasht, IR

Sep. 2015 - Sep. 2019

- **Thesis:** *Comparative Analysis of Modulation Methods in Visible Light Communication Systems*
- **Advisor:** Prof. Gholamreza Baghersalimi

Research Experience

Intelligent Maintenance and Operations Systems (IMOS) Lab. | EPFL

Doctoral Research Assistant (**Supervisor:** Prof. Olga Fink)

Lausanne, CH

Feb. 2023 - Present

- Advancing graph-based and physics-informed learning methods for computational sensing in networked dynamical systems.
- Investigating graph signal processing problems including graph inference, signal separation, sampling, and joint time-vertex analysis.
- Formulating unsupervised domain adaptation methods for graph-structured time series under distribution shifts.
- Establishing sensor virtualization and uncertainty quantification frameworks for reliable inference from imperfect measurements.

Data, Vibration and Uncertainty (DVU) Group | University of Cambridge

Visiting Researcher (**Supervisor:** Prof. Alice Cicirello)

Cambridge, UK

Sep. 2025 - Dec. 2025

- Proposed conformal prediction methods for uncertainty quantification in graph-structured time series.
- Introduced in-context learning frameworks using tabular foundation models for distribution-free predictive inference.
- Explored uncertainty-aware learning for reliable anomaly detection and sensing in dynamical systems.

Learning and Decisions Lab. | Aalborg University

Visiting Researcher (**Supervisors:** Prof. Rafael Wisniewski and Prof. Carsten Skovmose Kallesøe)

Aalborg, DK

May. 2024 - Jun. 2024

- Collected and analyzed multivariate sensor data from pipeline networks at the [Smart Water Infrastructures Laboratory \(SWIL\)](#).
- Investigated data-driven modeling, sensor virtualization, and smart-meter calibration for intelligent distribution networks.

Wireless Communications and Signal Processing (WCSP) Lab. | NTHU

Research Assistant (**Supervisor:** Prof. Chong-Yung Chi)

Hsinchu, TW

Sep. 2020 - Dec. 2022

- Proposed unsupervised methods combining robust statistics and deep learning for inverse problems in imaging.
- Applied convex optimization techniques to problems in machine learning, signal processing, and communications.

PranaQ

Machine Learning Research Engineer Intern (**Mentor:** Prof. Hau-Tieng Wu)

Taipei City, TW

May. 2022 - Aug. 2022

- Applied multimodal biomedical signal processing to SpO₂, blood pressure, pulse, and respiratory signals.
- Collaborated with physicians at [Taipei Medical University Hospital](#) on the collection and analysis of PPG, ECG, EMG, and EEG data.

Publications

Manuscripts Under Review:

- [1] Zepeng Zhang, Fuad Khuri, **Keivan Faghih Niresi**, and Olga Fink
[GSLAD: Prototype-Regularized Graph Structure Learning for Multivariate Time Series Anomaly Detection](#)
Submitted, 2026

Journal Papers:

- [1] **Keivan Faghih Niresi**, Christian Møller Jensen, Carsten Skovmose Kallesøe, Rafael Wisniewski, and Olga Fink
[Virtual Smart Metering in District Heating Networks via Heterogeneous Spatial-Temporal Graph Neural Networks](#)
Energy and Buildings, 2026

- [2] **Keivan Faghieh Niresi**, Alice Cicirello, and Olga Fink
In-Context Residual Calibration for Uncertainty Quantification of Energy Time Series over Graphs
Transactions on Machine Learning Research (TMLR), 2026
- [3] **Keivan Faghieh Niresi**, Dorina Thanou, and Olga Fink
Graph Signal Separation with Learnable Spectral Filters
IEEE Signal Processing Letters, 2026
- [4] Olga Fink, Ismail Nejjar, Vinay Sharma, **Keivan Faghieh Niresi**, Han Sun, Hao Dong, Chenghao Xu, Amaury Wei, Arthur Bizzi, Raffael Theiler, Yuan Tian, Leandro Von Krannichfeldt, Zhan Ma, Sergei Garmaev, Zepeng Zhang, Mengjie Zhao, Kevin Steiner, Yusuf Kesmen
From Physics to Machine Learning and Back: Part II—Learning and Observational Bias in Prognostics and Health Management (PHM)
Reliability Engineering & System Safety, 2026
- [5] Olga Fink, Vinay Sharma, Ismail Nejjar, Leandro Von Krannichfeldt, Sergei Garmaev, Zepeng Zhang, Amaury Wei, Gaetan Frusque, Florent Forest, Mengjie Zhao, Chi-Ching Hsu, **Keivan Faghieh Niresi**, Han Sun, Hao Dong, Chenghao Xu, Raffael Theiler, Arthur Bizzi, Kevin Steiner
From Physics to Machine Learning and Back: Part I—Learning with Inductive Biases in Prognostics and Health Management (PHM)
Reliability Engineering & System Safety, 2026
- [6] **Keivan Faghieh Niresi**, Jun Qing, Mengjie Zhao, and Olga Fink
Time-Vertex Machine Learning for Optimal Sensor Placement in Temporal Graph Signals: Applications in Structural Health Monitoring
Reliability Engineering & System Safety, 2026
- [7] **Keivan Faghieh Niresi**, Zepeng Zhang, and Olga Fink
RINS-T: Robust Implicit Neural Solvers for Time-Series Linear Inverse Problems
IEEE Transactions on Instrumentation and Measurement, 2025
- [8] **Keivan Faghieh Niresi**, Ismail Nejjar, and Olga Fink
Efficient Unsupervised Domain Adaptation Regression for Spatial-Temporal Sensor Fusion
IEEE Internet of Things Journal, 2025
- [9] **Keivan Faghieh Niresi**, Hugo Bissig, Henri Baumann, and Olga Fink
Physics-Enhanced Graph Neural Networks for Soft Sensing in Industrial Internet of Things
IEEE Internet of Things Journal, 2024
- [10] **Keivan Faghieh Niresi**, and Chong-Yung Chi
Robust Hyperspectral Inpainting via Low-Rank Regularized Untrained Convolutional Neural Network
IEEE Geoscience and Remote Sensing Letters, 2023
- [11] **Keivan Faghieh Niresi**, and Chong-Yung Chi
Unsupervised Hyperspectral Denoising Based on Deep Image Prior and Least Favorable Distribution
IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing, 2022

Peer-Reviewed Conference Papers:

- [1] **Keivan Faghieh Niresi**, and Olga Fink
Sensor Fault Detection via Virtual Smart Heat Metering with Spatial-Temporal Graph Neural Networks
European Conference of the Prognostics and Health Management Society (PHME), 2026
- [2] **Keivan Faghieh Niresi**, Lucas Kuhn, Gaëtan Frusque, and Olga Fink
Informed Graph Learning by Domain Knowledge Injection and Smooth Graph Signal Representation
European Signal Processing Conference (EUSIPCO), 2024
- [3] **Keivan Faghieh Niresi**, Mengjie Zhao, Hugo Bissig, Henri Baumann, and Olga Fink
Spatial-Temporal Graph Attention Fuser for Calibration in IoT Air Pollution Monitoring Systems
IEEE SENSORS, 2023

Invited Talks

- [1] Graph Deep Learning for Complex Systems Monitoring
CERN, Geneva, Switzerland, 2026
- [2] Graph Neural Networks for Smart Meters Monitoring
37th EURAMET TC Flow Meeting, Villeurbanne, France, 2026
- [3] Conformalizing Spatial-Temporal Graph Neural Networks with In-Context Learning
Dynamics and Vibration Tea Time Talks, University of Cambridge, Cambridge, United Kingdom, 2025
- [4] Physics-Guided Graph Inference for District Heating Networks
Fourth Workshop on Physics-Enhancing Machine Learning, Institute of Physics, London, United Kingdom, 2025
- [5] Physics-Enhanced Neural Networks for Energy Systems
Let's Talk Research, Lauzhack, Lausanne, Switzerland, 2025
- [6] Computational Sensing for Infrastructure Through the Lens of Graph Machine Learning
9th Intelligent Maintenance Conference, Lausanne, Switzerland, 2025

[7] [Graph Neural Networks for Environmental and Infrastructure Sensing](#)

Federal Institute of Metrology (METAS), Bern, Switzerland, 2024

[8] [Integrating Physics in Graph Neural Networks for Interaction Modeling](#)

Second Workshop on Physics-Enhancing Machine Learning in Applied Mechanics, Institute of Physics, London, United Kingdom, 2023

Teaching Experience

École Polytechnique Fédérale de Lausanne (EPFL)

Teaching Assistant

Lausanne, CH

Feb. 2023 - Present

- Data Science for Infrastructure Condition Monitoring (Spring 2026, Spring 2025, Spring 2024, Spring 2023)
- Machine Learning for Predictive Maintenance Applications (Fall 2026, Fall 2024, Fall 2023)
- Analysis II (Spring 2025)
- Analysis I (Fall 2024)

National Tsing Hua University (NTHU)

Teaching Assistant

Hsinchu, TW

Feb. 2021 - Jun. 2022

- Introduction to Convex Optimization (Spring 2022, and Spring 2021)

University of Guilan

Teaching Assistant

Rasht, IR

Feb. 2018 - Jun. 2019

- Electrical Circuits I (Spring 2019, Fall 2018, and Spring 2018)
- Communications Circuits (Spring 2019)

Skills and Expertise

Programming and Scripting: Python, MATLAB®, C, $\text{\LaTeX}$

Machine Learning and Data Science: PyTorch, PyTorch Geometric, SciPy, CVXPY, scikit-learn, pandas, NumPy, TensorFlow

Computer Vision and Imaging: OpenCV, scikit-image, Pyxu, DeepInverse, Kornia, SCICO

Data Infrastructure and DevOps: SQL, Git, Docker, Linux

Honors and Awards

- **EPFL EDCE Mobility Award**, *provided to selected PhD students to undertake an academic visit at an external research institution, 2025*
- **Travel Grant**, *awarded by Physics-Enhancing Machine Learning Workshop, 2025*
- **Awarded M.Sc. Full Scholarship (Merit-Based)**, *the highest award offered to NTHU graduate students, 2020*
- **Awarded B.Sc. Full Tuition-Waiving Scholarship**, *the highest award offered to Iranian undergraduate students, 2015*

Academic Services

Reviewer:

- **Journals:** IEEE Transactions on Network Science and Engineering, IEEE Transactions on Industrial Informatics, IEEE Internet of Things Journal, IEEE Signal Processing Letters, Mechanical Systems and Signal Processing, Artificial Intelligence Review, Engineering Applications of Artificial Intelligence, Internet of Things, IEEE Sensors Journal, Signal, Image and Video Processing
- **Conferences:** IEEE International Conference on Acoustics, Speech, and Signal Processing (ICASSP), IEEE International Workshop on Machine Learning for Signal Processing (MLSP), Learning on Graphs Conference, European Conference of the Prognostics and Health Management Society (PHME), International Joint Conference on Neural Networks (IJCNN)

Student Supervision (Co-advised with Prof. Olga Fink at EPFL):

- Hugo Walter, *Graph Neural Controlled Differential Equations for Long-Horizon Surrogate Modelling of Thermo-Hydro-Mechanical Repository Dynamics*, Internship and Master's thesis
- Matya Aydin, *Spatial-Temporal Denoising Diffusion Probabilistic Model for Industrial IoTs*, Semester project
- Julien Mollard, *Graph Neural Networks for Early Breast Cancer Detection*, Master's thesis
- Jun Qing, *Graph-Based Near-Optimal Sensor Placement: From Signal Processing to Neural Networks*, Master's thesis
- Lucas Kuhn, *Physics-Inspired Graph Signal Processing for District Heating Networks*, Semester project

Conference Organizer:

[Intelligent Maintenance Conference \(IMC\)](#) (2026, 2025, 2024, 2023)